
Week 7: Major Project - SkillSync Setup and Architecture

Week 7 marked the crucial initiation of the major project: **SkillSync**. This phase focused on synthesizing all previous learnings into a cohesive, production-ready AI-powered system.

Project Overview: SkillSync

SkillSync is designed as an intelligent platform for the synchronization and structured, multi-agent execution of complex projects or learning paths. It uses the CREW-AI framework as its core engine for coordination.

Architectural Blueprint

The focus was on a **three-tier architecture** for robustness and scalability:

- **Frontend (UI/UX):** A Streamlit or Flask-based interactive interface was designed to serve as the **User Dashboard** for input, workflow management, and result visualization.
- **Backend (Logic):** The core **Python-driven logic** integrated the **CREW-AI agents** for task planning and coordination, alongside direct **OpenAI API** calls for generating automated, intelligent suggestions and detailed step-by-step guidance.
- **Database (Persistence):** **MongoDB** was integrated to provide persistent data storage for user profiles, ongoing workflow states, and the long-term saving of generated project artifacts.

Key Development Challenges

The early development phase involved overcoming significant technical hurdles, including designing reliable **inter-agent communication** protocols, managing complex **asynchronous workflows** where agents operate in parallel, and ensuring robust, secure **API integration** across all services.